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PRACTICE MANAGEMENT

CRYSTAL PM

HOW-TO GUIDE

Set up a Crystal Meet room

Put a Crystal Meet device in a conference room and join Zoom and Google Meet calls from the room's screen, step by step.

This guide takes one conference room from an empty Raspberry Pi to a room that shows up in the Crystal Meet dashboard and joins calls from its own web page. Plan about 45 minutes per room, most of it waiting for downloads. No programming experience is needed: if you can follow a recipe and type the commands printed here, you can do this. Repeat it for each room.

STEP 1
Prepare the device
Install Raspberry Pi OS, sign in, connect the TV and camera.

STEP 2
Create the room
Add the room on the dashboard's **Provisioning** page and copy its token.

STEP 3
Install Crystal Meet
One command with the room's config file.

STEP 4
First run and screens
Join a test call, pick the TV's screensaver, then point the table screen and the door sign at the device.

BEFORE YOU BEGIN

Hardware per room. Raspberry Pi 4 (4 GB) or Pi 5, the official power supply, a 32 GB microSD card, an HDMI cable to the room's TV, and a USB camera and speakerphone (for example a Logitech Brio 505 and a Jabra Speak2 55).

Two screens on the network. A touch tablet or PoE touch panel on the table to join and control meetings, and a small screen by the door to show whether the room is free. Neither connects to the Pi: each only needs a browser, power, and the office network. Give the room device a fixed address on your router (a DHCP reservation) so the screens can always find it.

The dashboard. The Crystal Meet dashboard must already be running on the office network; the guide "Set up the Crystal Meet dashboard" installs it on its own Raspberry Pi with a fixed address, for example `http://10.0.0.50`. Rooms talk to it on port `3001`.

Accounts. A dashboard sign-in with the admin role, and the Zoom or Google Meet links you want to test with.

1 Prepare the device

- **Flash the card.** On your computer, open **Raspberry Pi Imager**, choose your Pi model, then **Raspberry Pi OS (64-bit)** with desktop, and your microSD card.
- **Pre-fill the settings.** When Imager offers customisation, set the hostname to something like `crystal-meet-room-1`, the username to `pi` with a password you keep, your Wi-Fi network and your time zone, and turn on SSH. Write the card.
- **Connect and boot.** Put the card in the Pi, plug in the TV over HDMI, the camera and speakerphone over USB, then power. Wait for the desktop to appear on the TV.
- **Turn on automatic sign-in.** Open **Preferences › Raspberry Pi Configuration › System** and set **Auto Login** to on, so the room comes back by itself after a power cut.

You should now see the Raspberry Pi desktop on the room's TV, and the Pi should be on the office network.

2 Create the room in the dashboard

- **Open the dashboard** at its address, for example `http://10.0.0.50` or `http://crystal-meet.local`, and sign in.
- **Go to Provisioning** and fill in the **Room Name** exactly as people call the room (for example `Room 1`) and, if you like, its **Location**. Press **Generate Token**.
- **Copy the token** shown under **Enrollment Token**. It is a long string of letters and numbers; you will paste it into the device's config in the next step.

You should now see the token on the page and, on **Devices**, the room listed as provisioning until its device connects.

> **GOOD TO KNOW**
>
> The token links one device to this room and works once. If a device is ever reinstalled, create a new token for it. Nothing else about the room needs to change.

3 Install Crystal Meet on the device

- **Open a terminal** on the Pi (the black icon in the top bar), or connect over SSH from your computer.
- **Get the software.** Type `sudo apt update && sudo apt install -y git`, then `git clone https://github.com/ben-abeo/croom.to.git`, then `cd croom.to`.
- **Make the room's config.** Type `cp deploy/rooms/room-1.yaml ~/room.yaml` (use `room-2.yaml` or `room-3.yaml` for the other rooms), then `nano ~/room.yaml`.
- **Fill in three values.** The room's **name** and, if people call the place something, its **location** (delete that line to show the name alone); the dashboard **url** as `http://<dashboard address>:3001`; and the **enrollment_token** you copied. Save with **Ctrl+O**, Enter, then **Ctrl+X**.
- **Run the installer.** Type `sudo bash installer/install.sh --config ~/room.yaml`. It installs the browser and the room software; expect 10 to 20 minutes.
- **Start it.** When the installer prints **Crystal Meet installation complete**, type `sudo systemctl start croom`. The service also starts on every boot from now on. If it ever stops, the **Start Crystal Meet** icon on the device's desktop (over VNC) starts it again.

  You should now see the room's screensaver fill the TV: the room's name, its status and the time. The room's TV needs no keyboard or mouse, and nobody has to touch it.

4 First run and the screens

- **Open the room page.** On your laptop or phone on the office network, go to `http://crystal-meet-room-1.local:8080`, or use the device's IP address with `:8080`.
- **Check the dashboard.** On **Devices**, the room should show **online** within a minute.
- **Join a test call.** Paste a Zoom link, or a Zoom meeting ID, into **Join with a link** and press **Join**. A meeting ID, or a Zoom link without its passcode, makes a **Passcode** field appear under the box: type the meeting's passcode there before you press **Join** (on the table screen, tapping the field opens the on-screen keyboard on its digits). The TV shows the meeting joining; the page switches to the meeting screen, which offers **Mute**, **Turn camera off** and **Leave** and, depending on the platform, more (see **The meeting screen** below).
- **Leave.** Press **Leave**, then **Tap again to leave**. The TV returns to the screensaver.
- **Pick the TV's idle look.** Tap **Options**, then **Screensaver** on the room page, and choose **Information** (name, status, next booking and the time), **Quiet** (name and status only), **Brand** (the logo) or **Bounce** (the logo drifting around the screen). The TV changes at once and the choice is remembered.
- **Connect the calendar.** To show real bookings and let people join them, follow the guide **Connect Crystal Meet rooms to Google Calendar** next. Until then the room page works with pasted links only.
- **Connect Zoom.** To join Zoom meetings, including ones hosted by other companies, follow the guide **Connect Crystal Meet rooms to Zoom**. Without it, Zoom turns the room away as an automated guest.

- **Set up the table screen.** On a Raspberry Pi with the touch display, clone the repository as in step 3 and type `sudo bash installer/install-kiosk.sh --url http://<device address>:8080/`. The room page opens full screen at every login with its own on-screen keyboard, and a **Room controls** icon on the desktop reopens it if it is ever closed; run the command again to change the address. After updating Crystal Meet, pull the repository on the table Pi too, run the install-kiosk command again and reboot it so the new scale takes effect. On a tablet, open its kiosk browser (Fully Kiosk Browser on Android, Guided Access with Safari on an iPad), set the start page to `http://<device address>:8080/`, turn on full screen and keep the screen awake.
- **Type on the table screen.** Tap the link box (or the passcode field) and the on-screen keyboard comes up from the bottom of the screen. It has three layers, like an iPad's: letters first, then **123** for numbers and the symbols a link or a passcode needs, then **#+=** for the rest; **ABC** goes back to the letters. Tap the shift key (the up arrow) once for one capital, or twice quickly to lock capitals on, as caps lock does, and a dot shows under its arrow; tap it again to let go. Every key answers the moment a finger lands, so both thumbs can type at once, and holding the delete key keeps deleting. Each key clicks softly, and a letter or symbol shows a larger copy of itself above the key while you press it. **Join** on the keyboard joins the meeting you typed; the key with the down arrow puts the keyboard away, and so does a tap anywhere else on the page. To turn the click off, tap **Options**, then **Key click off** under **Keyboard**; **Key click on** brings it back. The room remembers the choice, for every screen that opens its page.
- **Set up the door sign.** On the door screen, set the start page to `http://<device address>:8080/sign`. It shows green when the room is free, amber when a meeting is about to start, and red while the room is in use or booked.
- **Set up the camera and sound.** Tap **Options**, then **Camera framing** (the TV shows the camera between meetings; a screen left alone returns to the idle screen after ten minutes, and **Back** returns at once), press **Find the stops**, nudge with the arrows and zoom, then **Save as Home**. Frame the table, press **Save**, then a preset; repeat for the whiteboard. Tap a preset to return to it, even in a call. Moves take seconds (**Moving…**); the camera goes Home at every service start. During a call the page shows the meeting screen on its own: **Mute**, **Turn camera off**, **Leave**, the volume, **Camera** for a preset or a nudge, and **Upcoming meetings** to see what is booked next. On that screen, under **Sound**, set the level and try **Mute speaker**; `audio.output_device: HDMI` in the room config (then restart) sends the sound to the TV. **No speaker found** or **No controllable camera found**: the device sees no speaker, or no camera that pans or zooms (the MeetUp does).

  You should now see the room page say the room is free again, the door sign green, and the dashboard still showing the device online.

> **TIP**
>
> Zoom links that include their passcode (the part after `?pwd=`) join straight through. Google Meet needs the room signed in once as its own Workspace user; the guide "Let Crystal Meet rooms join Google Meet" covers it.

ON THE TABLE

The meeting screen

While a call lasts the page shows the meeting screen on its own, and draws only what the call's platform offers the room right now. This is the screen at its fullest, in a Zoom meeting where the room is the host. A Google Meet call offers Mute, Turn camera off, Raise hand, Captions and Leave, and no more. Every button has an icon before its label (a dialog's card, below, is the exception); where a button shows a state, the icon is the state and the words are what a tap does.

The first row

Mute or **Unmute**
The microphone as the meeting has it: a host muting the room changes the button within a few seconds.

Turn camera off or **Turn camera on**
The same for the camera.

Raise hand or **Lower hand**
Zoom keeps the hand in its own Reactions list; the room presses it there.

Reactions
Opens a row for ten seconds: Thumbs up, Clap, Heart, Laugh, Wow, Celebrate, and Cancel.

Leave
Asks for a second tap: Tap again to leave.

The second row

Captions on or **Captions off**
Zoom's captions on the TV.

Gallery view or **Speaker view**
The button names the view a tap switches to.

Hide self view or **Show self view**
The room's own picture on the TV.

Participants (7)
Opens the list of people; the count leaves out the waiting room.

Chat (2)
Opens the chat; the number is the messages not yet read.

Host
Only while the room is host or co-host.

The screens behind the second row

Participants
Everyone in the meeting, marked Host, Hand raised, Muted or Video off; the room says "(this room)". A host or co-host also gets Mute and Remove on each other person (Tap again to remove), and, under Waiting room, Admit for each person waiting.

Chat

The last 20 messages other people sent, newest at the bottom, with sender and time; No messages yet when there are none. It is read-only: there is no reply box.

Host controls

Mute all, Ask all to unmute, Start recording or Stop recording, Lock meeting or Unlock meeting, and, for the host only, End meeting for everyone (Tap again to end).

Each of the three has **Back**, and each goes back to the meeting screen on its own after two idle minutes; Camera and Upcoming meetings stay where you leave them. A tap anywhere on the screen starts the two minutes again.

What the meeting says to the room

- **The notice line.** A line under the meeting's title says what the meeting has disclosed of itself (recorded, transcribed, AI Companion) and stays for the rest of the call.
- **The dialog card.** A Zoom dialog that waits for a tap is a card above the controls, with Zoom's own words and buttons in Zoom's order, the last one highlighted (Zoom puts the cancelling button first). The card's buttons carry no icons, because the words are Zoom's; its buttons ignore a tap for half a second after it appears, so a tap meant for the controls under it answers nothing by mistake. The Zoom guide says which dialogs wait and which are pressed for the room.
- **Second taps.** Leave, Remove and End meeting for everyone ask for a second tap: Tap again to leave, Tap again to remove, Tap again to end. Each must come within five seconds, and a double tap on the glass does not count as the second.

Icons are from Lucide (lucide.dev), ISC licence: the symbols above are the ones the table draws.

IF SOMETHING IS OFF

Troubleshooting

The room page does not load

On the Pi, type `sudo systemctl status croom`. If it is not running, type `sudo journalctl -u croom -n 50` and look for the first line marked ERROR. Make sure port 8080 is not used by something else on the device.

The device shows offline

From the Pi, type `curl http://<dashboard address>:3001/health`. If nothing comes back, the dashboard is not reachable from the room's network: check the address in `~/room.yaml` and that the dashboard Pi is up (its guide has the checks). If the token was already used, create a new one and run the installer again with the updated config.

The browser opens but never joins

Zoom needs the meeting's passcode: paste the full invitation link, or type the passcode in the **Passcode** field that appears for a meeting ID or a link without one (on a "Couldn't join" card that says "Zoom needs the meeting passcode", press **Join again**). Google Meet needs the room signed in (the guide "Let Crystal Meet rooms join Google Meet"); the room page says so when the sign-in is missing or expired. The room page quotes what Meet showed and the log names a screenshot under `/tmp`; from the Pi, `PLAYWRIGHT_BROWSERS_PATH=/opt/croom/browsers DISPLAY=:0 /opt/croom/venv/bin/croom --check-meet <link>` reports the same (stop the service first: `sudo systemctl stop croom`).

No sound or picture in the call

Check the USB camera and speakerphone are connected before the device starts, then reboot the Pi. The service runs as the signed-in desktop user so it can use the room's screen and audio. From the Pi, `PLAYWRIGHT_BROWSERS_PATH=/opt/croom/browsers DISPLAY=:0 /opt/croom/venv/bin/croom --check-meet <link>` (stop the service first: `sudo systemctl stop croom`) prints "Devices the browser sees" with the microphones, speakers and cameras the browser can use; if the MeetUp is missing from "Devices the browser sees" but `wpctl status` lists it, the browser cannot reach PipeWire: make sure `libpulse0` is installed (the installer adds it) and that the service runs in the desktop user's session.

No browser window opens on the TV

The service waits for the desktop before it starts, so give a freshly booted Pi a minute. If the screensaver still does not appear, type `sudo journalctl -u croom -n 50` and look for "Failed to initialize". The browser is installed under `/opt/croom/browsers`; if that folder is missing, run the installer again.

The screens cannot open the page

Use the device's IP address rather than its `.local` name; many tablets do not resolve those. Check that the screen and the device are on the same network, then open `http://<device address>:8080/api/status` from a laptop to confirm the device answers.

GOOD TO KNOW

Under the hood the device software is still called `croom` : that is the name you will see in service and log commands. Everything people see says Crystal Meet.